

What ***Exactly*** is a Fractal?

Mathematics is the art of precision. So why is this term so tough to pin down?



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Trees and clouds are wonderful examples of fractal structure in Nature.
(Photo by Author)

“In science, all important ideas need names and stories to fix them in the memory.” — Benoit B. Mandelbrot (1924–2010)

Major milestones in mathematics are generally measured

in centuries. We are fortunate to come of age in the wake of a fundamentally new perspective in mathematical thought.

In 1975, Benoit Mandelbrot published *Les Objets Fractals*. The book described the common thread of scaling symmetry that united the untamed forms of the natural world with the work of notable mathematicians like Weierstrass, Sierpinski, Koch, Hausdorff, and Cantor (see links below). Over the previous century, these great thinkers had explored functions that were then considered to be “pathological” — shapes like the Weierstrauss function, which is continuous yet nowhere differentiable.

Through an amazing and circuitous path, in 2002 I suddenly found myself working with Benoit while he was at Yale University. His lectures were breathtaking in scope. Benoit's knowledge was encyclopedic — he often connected disparate fields in fascinating ways. He also enjoyed sharing the origin story of how he came to recognize fractal geometry and how his thinking on the subject evolved.

Benoit's Insight

At first, his idea of a new kind of geometry that included physical dimensions that are “in-between” the standard Euclidean dimensions was met with skepticism. Over time, however, the language of fractal geometry has proved to be an invaluable tool for both analyzing and describing an incredible variety of natural phenomena.

One of the reasons fractal geometry has gained popularity over the years is that the idea of self-similarity is intuitive and easy to grasp — a fractal object is, in some sense, composed of smaller copies of itself. But fractal geometry is more than just self-similarity and, as a result, the subject is also very easy to

misunderstand. In the rush to search for fractal characteristics, even research scientists have made errors by overlooking essential principles.

So, what precisely did Benoit mean by the term “fractal”?

The Hausdorff Dimension

In his seminal book from 1982 entitled *The Fractal Geometry of Nature*, he stated:

“A fractal is by definition a set for which the Hausdorff-Besicovitch dimension strictly exceeds the topological dimension.”

Broadly speaking, the Hausdorff-Besicovitch dimension, or simply “Hausdorff dimension” is a measure of roughness.

Often abbreviated as $\dim_H$, it serves as a metric for how an object (or set) scales.

While the derivation is deep and the calculations can be involved, for a simple example of how it works, we can look at

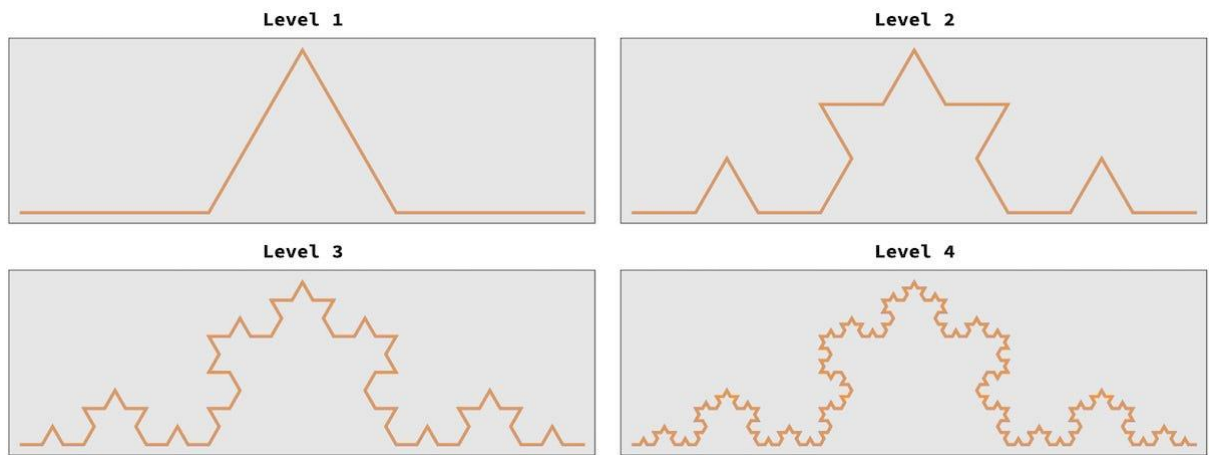
the *Koch curve*. It is named after Swedish mathematician Neils Fabian Helge von Koch (1870–1924) who, at the time of inventing the shape, had been studying continuous, non-differentiable curves. This curve was influential in Mandelbrot's thinking.

Figure 1 shows the generation of the shape. Starting with a straight line, in Level 1 we:

- 1) remove its middle third, and
- 2) replace it with a tent shape made from two segments that are each $\frac{1}{3}$ the length of the line.

To obtain the next level of scaling, we simply replace each straight line in the current level by repeating the above Steps 1 and 2.

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First four levels of the Koch curve. (Image by Author)

Because it is constructed from 4 copies that are each scaled by a factor of $1/3$, we can quickly calculate its Hausdorff dimension using the relation for the similarity dimension d ,

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$$d = \frac{\text{Log}(N(s))}{\text{Log}(1/s)}$$

which gives us:

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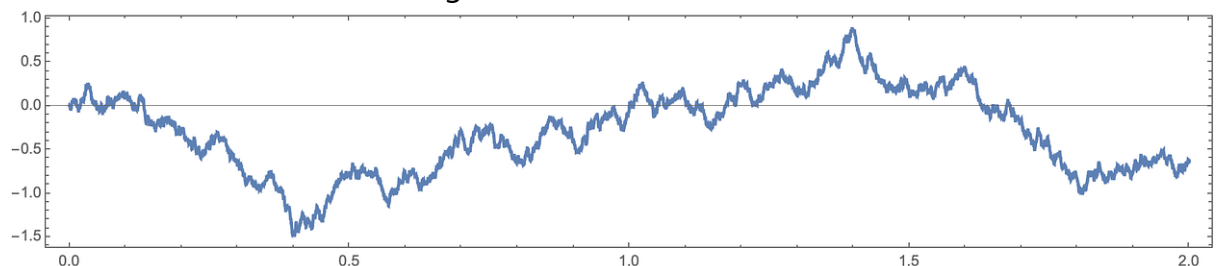
$$\dim_H = \frac{\log(4)}{\log(3)} \approx 1.26186.$$

The dimension tells us that the shape is more than a simple line and quantifies its propensity to fill the plane. Importantly, what $\dim_H$ tells us is that attempting to measure the curve as a 1-dimensional line would give us an infinite length, while

attempting to measure it as a planar object would give us an area equal to zero. The Hausdorff dimension is therefore the appropriate measure.

Now for the problem with Benoit's early definition in which the Hausdorff dimension strictly exceeds the topological dimension: there are fractals for which the Hausdorff dimension *equals* the topological dimension. For example, the Takagi-Blancmange curve and fractional Brownian motion with a Hurst index of $1/2$ are both fractal, 1-dimensional curves with a $\dim_H = 1$.

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An example of fractional Brownian motion with Hurst exponent $1/2$ and $\dim_H=1$. (Image by Author)

A Little Knowledge is a Dangerous Thing

As a result of these types of examples, Benoit realized that his original definition was overly restrictive. Here is a later, more accommodating description of his:

“A fractal is a rough or fragmented geometric shape that can be split into parts, each of which is (at least approximately) a reduced-size copy of the whole.”

Predictably, many folks never got the message. Contributing to the confusion is that the internet is littered with incomplete and misleading characterizations. Over the years, the combination of conceptual simplicity, ubiquity, and deep applications has led to conflicting notions of what constitutes fractal structure.

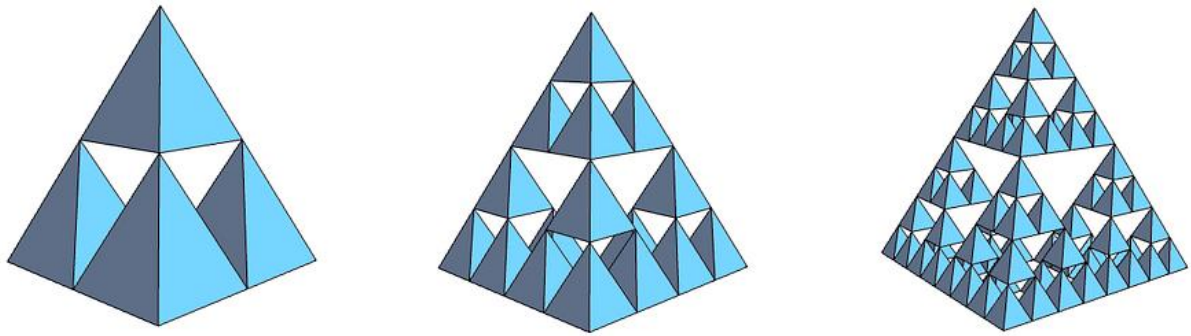
Often, this misunderstanding involves a focus on the idea that, in principle, you can continue zooming into a fractal *forever* to find more of the same structure. It is true that for purely mathematical constructions, the only scaling limitation is computational power. However, the universe is filled with fractal structure that manifests over a range that is limited by physical constraints. In Benoit’s eyes, mountains, lightning, clouds, and broccoli were solid examples of real fractals, not lowly cousins of their famous mathematical brethren.

There is also a related notion that fractals are perfectly self-similar. While many are, Nature alone shows us that this is not a necessary condition. Even the Mandelbrot set itself is not perfectly self-similar!

Another fundamental misunderstanding arises from the idea that self-similarity is synonymous with fractality. Self-similarity is a necessary but insufficient condition for claiming fractal structure. For example, onions, spirals, and Russian dolls are not fractal — they are self-similar only around a point. “Onions” might sound like a silly example, but years ago somebody published a book about fractals and music claiming that onions are fractal. The worst part is that I've seen the book cited in academic research.

Finally, the notion that what makes something fractal is the fact of having a non-integer dimension is just plain wrong. My favorite example of this is the Sierpinski tetrahedron. It is constructed from four tetrahedra each with base lengths that are scaled by a factor of $1/2$:

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The first three levels of the Sierpinski tetrahedron. (Image by Author)

Here is the dimension calculation:

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$$\dim_H = \frac{\log(4)}{\log(2)} = 2.$$

No mathematician would claim that this archetypal structure is not a fractal.

Basic Requirements for Fractal Structure

In preparation for the NSF-funded, 2004 Yale Fractal Geometry Workshop, Michael Frame and I wanted to present a clear set of criteria for understanding what qualifies an object as fractal in the most basic sense.



Left to right: Michael Frame, the author, and Benoit Mandelbrot at a Yale Fractal Geometry Workshop.

What follows are the requirements we arrived at.

Fractals are *self-similar with respect to scale*, by which we mean that any such object satisfies the following three requirements:

It is constructed of a collection of different-sized elements whose size distribution satisfies a power-law relationship spanning at least three scales:

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Not fractal

A minimum of three levels of scaling is a necessary condition. (Image by Author)

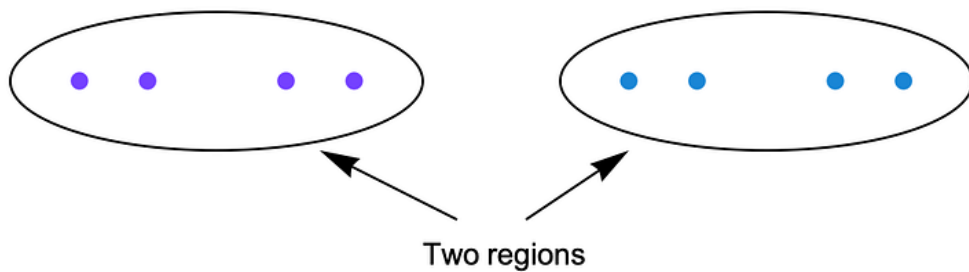


Possibly fractal



2. It must comprise at least two similar regions in which the arrangement of elements either mirrors or imitates the structure of the object as a whole:

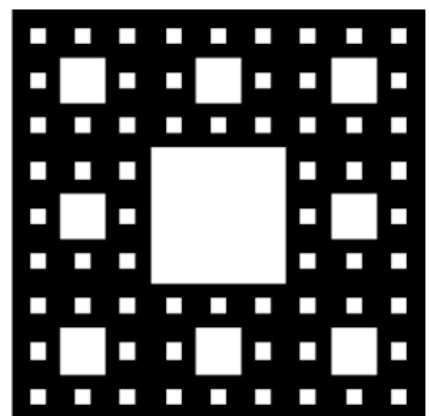
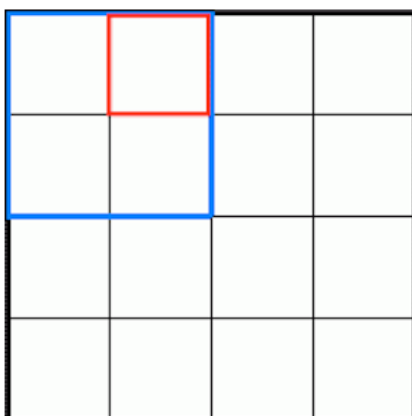
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Scaling must appear in at least two regions. (Image by Author)

3. Its features must possess sufficient detail that the overall structure cannot be more easily explained in Euclidean terms:

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The left-hand figure is not fractal. The right-hand figure is fractal known as a Sierpinski carpet. (Image by Author)

Although the blue and red outlines in the left-hand figure above mark smaller copies of the large square, the overall shape can be adequately described in terms of an evenly subdivided square; specifically, it is not *necessary* to invoke the words “scaled” or “recursive” to explain, in a concise manner, how it is constructed. The structure of the Sierpinski carpet on the right, while also composed of scaled squares, clearly requires a more complicated description.

Coda

While the above criteria are oriented toward physical objects, the same concept applies to statistical self-similarity. Make no mistake, fractal geometry gets very deep, especially when applied to more abstract realms like time series and number theory. When more than one power-law is at play, the simple rules are no longer sufficient, and we need to turn to multifractal analysis.

As for Benoit’s definition of a fractal, he was a true scientist, unafraid to update his perspective when presented with new information. In fact, there’s a wonderful story that Michael

Frame shared about when he began collaborating with Benoit. Michael had been working on a difficult problem and arrived at results that contradicted Benoit's expectations. He was concerned about what his reaction might be. Benoit's response?

“Marvelous, the problem is more interesting than I thought.”

Down the Rabbit Hole...

Michael Frame's website at Yale: [Fractal Geometry](#)

MacTutor biography: [Karl Theodor Wilhelm Weierstrass](#)

MacTutor biography: [Wacław Sierpiński](#)

MacTutor biography: [Niels Fabian Helge von Koch](#)

MacTutor biography: [Felix Hausdorff](#)

MacTutor biography: [Georg Ferdinand Ludwig Philipp Cantor](#)

A formal definition of the Hausdorff dimension: [Hausdorff dimension](#)